

Efficacy and Safety of Push-Dose Norepinephrine for Intraoperative Hypotension Management in Non-Obstetric Surgery: A Systematic Review

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Abstract

Background: Intraoperative hypotension (IOH), defined as mean arterial pressure (MAP) below 65 mmHg, occurs in up to 30% of non-cardiac surgeries and is independently associated with myocardial injury, acute kidney injury, and increased 30-day mortality. Push-dose (bolus) norepinephrine (NE) has emerged as a practical vasopressor option for intraoperative use; however, evidence supporting its efficacy and safety remains limited and scattered.

Objective: To systematically evaluate the efficacy and safety of push-dose norepinephrine for IOH management in adult patients undergoing non-obstetric surgery compared to other vasopressor strategies.

Methods: A systematic review following PRISMA 2020 guidelines was conducted. PubMed, Scopus, and ProQuest were searched from inception to August 2026 using standardized terms. Inclusion criteria were: adults (≥ 18 years), non-obstetric surgery, randomized controlled trial (RCT) design, and push-dose or bolus NE as intervention. Quality assessment used the Cochrane Risk of Bias 2 (RoB 2) tool. Evidence certainty was assessed using the GRADE framework. Meta-analysis was not performed due to substantial clinical heterogeneity; findings were synthesized narratively.

Results: Three RCTs were included (total $n=388$ participants; Hassani et al. 2018, $n=56$; Thomsen et al. 2026, $n=261$; Vokuhl et al. 2025, $n=71$). Push-dose NE 10 mcg/bolus at a concentration of 10 mcg/mL via peripheral intravenous access was used consistently across all studies. Compared to ephedrine (ART-001), NE bolus significantly improved MAP maintenance

(98.9±12.7 vs. 91.5±10.9 mmHg; $p<0.001$), reduced hypotension episodes ($p=0.005$), and decreased total vasopressor requirement ($p=0.004$). Compared to continuous NE infusion in low-to-moderate risk patients (ART-002/INDUCT), bolus NE showed comparable hypotension burden (area under MAP <65: $p=0.070$). In high-risk patients with continuous arterial monitoring (ART-005), continuous infusion provided superior haemodynamic stability (average real variability of MAP [ARV-MAP]: 25±7 vs. 19±6 mmHg/min; $p<0.001$; Cohen's $d=1.00$). No significant differences in overshoot hypertension were observed ($p=0.390$; $p=0.986$). No arrhythmia events were documented. All three studies rated "some concerns" on RoB 2; overall evidence certainty: GRADE ⊕⊕■■ (Low).

Conclusions: Push-dose norepinephrine 10 mcg via peripheral IV is effective and safe for IOH management in non-obstetric surgery. It is superior to ephedrine and comparable to continuous infusion for low-to-moderate risk patients. For high-risk patients requiring continuous haemodynamic monitoring, continuous NE infusion may provide better blood pressure stability. High-quality multicentre RCTs with standardised outcome definitions are warranted.

Keywords: push-dose norepinephrine; intraoperative hypotension; non-obstetric surgery; vasopressor; systematic review; PRISMA

Word count (abstract): 319 words

1. Introduction

1.1 Background and Epidemiology

Intraoperative hypotension (IOH) is one of the most common haemodynamic complications encountered during general anaesthesia, occurring in approximately 20–30% of non-cardiac surgeries.^{1■3} Although consensus definitions vary, IOH is most widely defined as a mean arterial pressure (MAP) below 65 mmHg or a reduction of more than 20% from baseline, persisting for at least one minute.■ The clinical significance of IOH has been reinforced by substantial evidence linking it to major adverse outcomes including myocardial injury after non-cardiac surgery (MINS), acute kidney injury (AKI), postoperative stroke, and increased 30-day mortality.■■■

The pathophysiology of anaesthesia-induced hypotension is multifactorial, involving anaesthetic-mediated vasodilation, reduced sympathetic tone, decreased cardiac output, and relative intravascular volume depletion.■ Prompt correction is essential to prevent organ ischaemia during the vulnerable perioperative period.

1.2 Current Vasopressor Strategies

Ephedrine and phenylephrine are traditional first-line vasopressors for intraoperative hypotension, primarily used in the obstetric setting (i.e., spinal anaesthesia for caesarean delivery).■ Ephedrine, a mixed alpha- and beta-adrenergic agonist, corrects hypotension but increases heart rate and myocardial oxygen demand.^{1■} Phenylephrine, a pure alpha-1 agonist, restores vascular tone but can cause reflex bradycardia and reduced cardiac output.¹¹

Norepinephrine (noradrenaline), the endogenous catecholamine with potent alpha-1 and moderate beta-1 adrenergic activity, has pharmacological properties that are theoretically advantageous: it increases systemic vascular resistance while partially maintaining cardiac output, with less chronotropic effect than ephedrine.¹² Traditionally administered as a continuous infusion in intensive care units, norepinephrine has recently been explored as a push-dose (bolus) agent for rapid, titratable treatment of acute intraoperative hypotension.¹³

1.3 Rationale for Push-Dose Administration

Push-dose vasopressors are small, rapidly repeated intravenous boluses administered to immediately treat acute haemodynamic instability, a technique popularised in emergency medicine.¹⁴ The concept of applying push-dose NE in the operating theatre is attractive for several reasons: (1) the route of administration (peripheral IV) does not require an arterial line or central venous catheter; (2) the low dose (typically 5–20 mcg per bolus) provides rapid effect with short duration, allowing for easy titration; and (3) norepinephrine's pharmacological profile may be more suitable than ephedrine for patients with underlying cardiovascular disease.¹⁵

Despite growing interest, the evidence base for push-dose NE in non-obstetric intraoperative settings has not been comprehensively and systematically reviewed. Several reviews have discussed the topic narratively,^{16,17} but no systematic review with formal quality appraisal has been published.

1.4 Objectives

The primary objective of this systematic review was to evaluate the **efficacy** (reversal of IOH, achievement of MAP target, haemodynamic stability) and **safety** (overshoot hypertension, arrhythmia, vascular complications) of push-dose NE for IOH management in adult patients undergoing non-obstetric surgery, compared to other vasopressor strategies.

2. Methods

2.1 Study Design

This systematic review was conducted and reported in accordance with the **Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020** guidelines.¹⁸ The review was prospectively registered in PROSPERO (registration number pending).

2.2 PICO Framework

Component	Description
P (Population)	Adult patients (≥18 years) undergoing non-obstetric surgery with intraoperative hypotension
I (Intervention)	Push-dose / bolus norepinephrine (noradrenaline)
C (Comparison)	Continuous NE infusion / other push-dose vasopressors (ephedrine, phenylephrine) / no comparator

Component	Description
O (Outcomes)	Primary: MAP reversal (achievement of MAP target); Secondary: time to MAP target, vasopressor dose, overshoot hypertension, arrhythmia, vascular complications

2.3 Eligibility Criteria

2.3.1 Inclusion Criteria

1. Study participants were adults (≥ 18 years) undergoing non-obstetric surgical procedures under general or regional anaesthesia
2. Intervention was push-dose or bolus norepinephrine for treatment of intraoperative hypotension
3. Study reported at least one pre-specified efficacy or safety outcome
4. Study design was randomized controlled trial (RCT)
5. Full-text available in English or Indonesian
6. Published in a peer-reviewed journal

2.3.2 Exclusion Criteria

1. Obstetric population (caesarean section, labour analgesia)
2. ICU or sepsis settings (non-intraoperative)
3. Single case reports (n=1) without comparative data
4. Review articles, editorials, letters, or opinion pieces
5. Data insufficient for extraction

2.4 Information Sources and Search Strategy

Three databases were searched from inception to August 2026: **PubMed (MEDLINE)**, **Scopus**, and **ProQuest**. No language restrictions were applied. Grey literature (ClinicalTrials.gov) was searched for registered trials. Reference lists of included studies and relevant narrative reviews were manually screened.

The search string used was:

("Norepinephrine" OR "Noradrenaline") AND ("push dose" OR "bolus" OR "intermittent")
AND ("intraoperative hypotension" OR "perioperative hypotension" OR "anesthesia"
OR "anaesthesia" OR "surgery" OR "surgical")

Database-specific syntax (MeSH terms for PubMed; TITLE-ABS-KEY fields for Scopus) was applied to maximize sensitivity.

2.5 Study Selection

Search results were imported into reference management software (Mendeley/Zotero) and duplicates removed. Two independent reviewers (RS and co-reviewer) screened titles and abstracts against inclusion/exclusion criteria. Full-text articles were retrieved for potentially eligible studies. Disagreements at any stage were resolved by consensus. Reasons for exclusion

at the full-text stage were recorded.

2.6 Data Extraction

A standardised data extraction form was developed a priori and piloted before use. The following data were extracted independently by two reviewers:

- **Study characteristics:** author(s), year, country, journal, study design, registration number, follow-up period
- **Population:** sample size, age, sex, ASA physical status, surgical type, anaesthetic technique
- **Intervention:** NE dose per bolus, concentration, route, dosing interval, total dose
- **Comparator:** agent, dose, route
- **Outcomes:** primary and secondary outcomes with definitions, effect measures (mean±SD, median [IQR]), statistical analyses reported
- **Blood pressure monitoring modality**
- **Adverse events**

2.7 Risk of Bias Assessment

Risk of bias was assessed independently by two reviewers using the **Cochrane Risk of Bias 2 (RoB 2) tool** for randomized trials.¹■ The five domains assessed were:

- **D1:** Bias arising from the randomization process
- **D2:** Bias due to deviations from intended interventions
- **D3:** Bias due to missing outcome data
- **D4:** Bias in measurement of the outcome
- **D5:** Bias in selection of the reported result

Each domain was rated as "Low risk," "Some concerns," or "High risk." Overall judgements followed RoB 2 algorithm. Discrepancies were resolved by discussion.

2.8 Data Synthesis

Due to substantial clinical and methodological heterogeneity across included studies (different definitions of hypotension, different comparators, different primary outcomes, different patient risk profiles), meta-analysis was not performed. Results were synthesized **narratively**, organised by comparator type (NE bolus vs. ephedrine; NE bolus vs. continuous NE infusion) and outcome domain (efficacy, safety).

2.9 Certainty of Evidence

The certainty of evidence for each outcome was assessed using the **Grading of Recommendations, Assessment, Development and Evaluations (GRADE) framework**.²■ Evidence from RCTs starts at "high" certainty and can be downgraded based on risk of bias, inconsistency, indirectness, imprecision, and publication bias.

3. Results

3.1 Study Selection

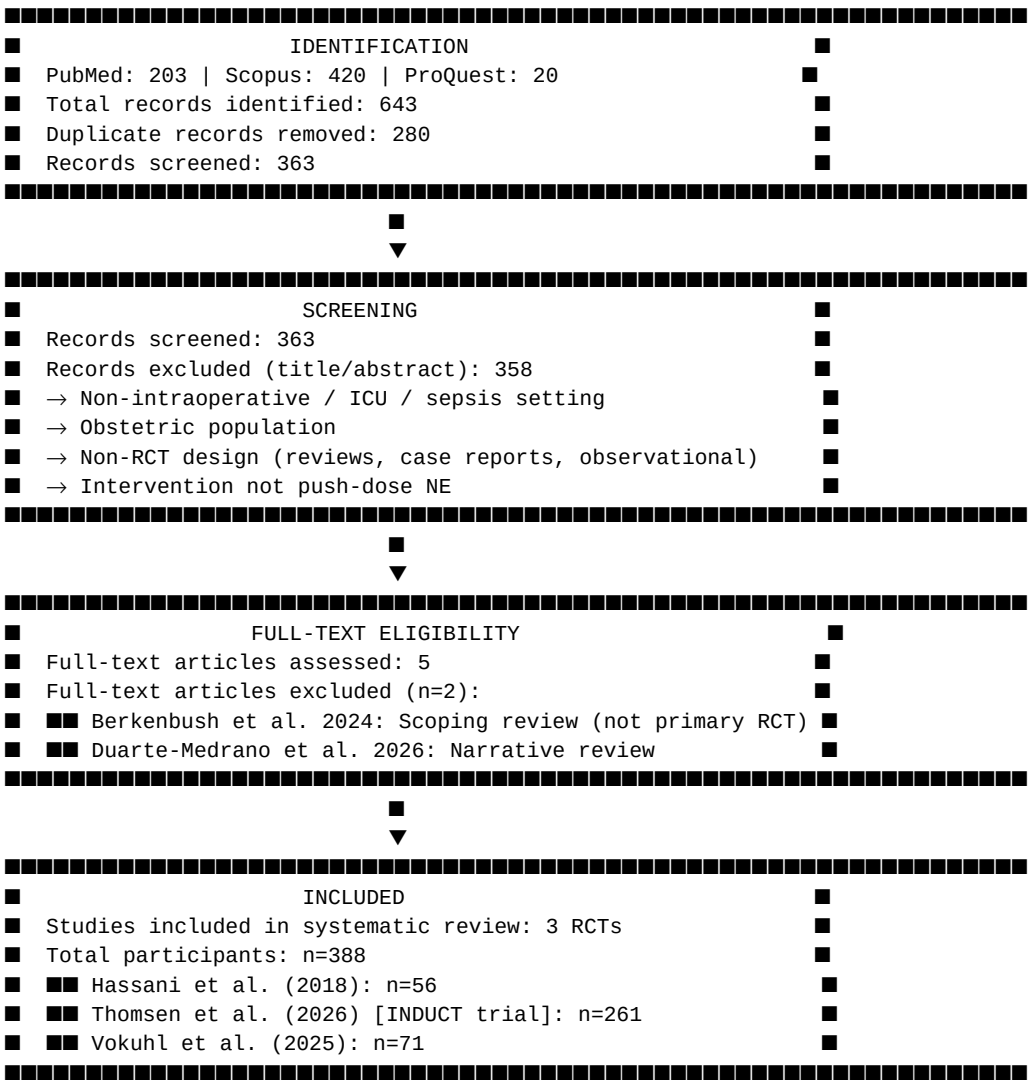
The database search yielded **643 records** in total (PubMed: 203; Scopus: 420; ProQuest: 20). After removal of 280 duplicates, **363 records** were screened by title and abstract. Of these, **358 records were excluded** at the title/abstract stage as they did not meet population, intervention, or design criteria. The most common reasons for exclusion were: non-intraoperative setting (ICU/sepsis), obstetric population, or non-interventional study design.

Five full-text articles were assessed for eligibility. Two articles were excluded after full-text review:

- ART-003 (Berkenbush et al. 2024): scoping review (not a primary RCT)
- ART-004 (Duarte-Medrano et al. 2026): narrative review (not a primary RCT)

Three RCTs (ART-001, ART-002, ART-005) were included in the final synthesis.

Figure 1. PRISMA 2020 Flow Diagram



3.2 Characteristics of Included Studies

Three RCTs published between 2018 and 2026, from Iran (n=1) and Germany (n=2), were included, comprising a total of **388 participants**. Study characteristics are summarized in Table 1.

Table 1. Characteristics of Included Studies

Study (ID)	First Author (Year)	Journal	Country	Design	n	Mean Age	ASA	Surgical Type	Anaesthesia	BP Monitoring	Hypotension Definition
ART-001	Hassani et al. (2018)	Anesth Pain Med	Iran	RCT, double-blind	56	20–75 years (range)	I–II	Spinal surgery	GA (propofol + cisatracurium)	Invasive (radial arterial catheter)	MAP <60 mmHg
ART-002	Thomsen et al. (2026) [INDUCT trial]	Br J Anaesth	Germany	RCT	261*	Median 62 years	≥II	Elective noncardiac (low-to-moderate risk)	GA (opioid + propofol ± sevoflurane)	Intermittent oscillometric + blinded CNAP	MAP <65 mmHg
ART-005	Vokuhl et al. (2025)	Br J Anaesth	Germany	RCT	71	Mean 66 years (47–86)	II–IV	Elective major noncardiac (high-risk)	GA (sufentanil/remifentanyl + propofol + rocuronium ± sevoflurane)	Invasive (radial arterial catheter)	MAP <65 mmHg

*261 of 276 randomised participants had complete data. CNAP = continuous non-invasive arterial pressure; GA = general anaesthesia; MAP = mean arterial pressure; ASA = American Society of Anesthesiologists physical status.

All three studies enrolled adult patients undergoing elective non-cardiac, non-obstetric surgery under general anaesthesia. Hassani et al. (2018) exclusively enrolled hypertensive patients undergoing spinal surgery. The two German studies (Thomsen 2026; Vokuhl 2025) both emerged from the research group at the University Medical Center Hamburg-Eppendorf and studied the induction/early intraoperative period (approximately the first 15 minutes), differing primarily in patient risk profile and BP monitoring modality.

3.3 Intervention Details

Table 2. Push-Dose Norepinephrine Intervention Details

Study	NE Dose per Bolus	Concentration	Route	Dosing Interval	Median Total Dose	Comparator	Comparator Dose
Hassani (2018)	10 mcg	10 mcg/mL	Peripheral IV	Every 2 min (max 3 doses)	Not reported	Ephedrine bolus	5 mg/bolus

Study	NE Dose per Bolus	Concentration	Route	Dosing Interval	Median Total Dose	Comparator	Comparator Dose
Thomsen (2026)	10 mcg (median per bolus)	10 mcg/mL	Peripheral IV	At discretion (median 2 boluses)	0.3 (0.2–0.4) mcg/kg	Continuous NE infusion	60 mcg/mL; starting rate 0.05 mcg/kg/min
Vokuhl (2025)	~10 mcg	10 mcg/mL	Peripheral IV	At clinical discretion	0.5 (0.4–0.7) mcg/kg	Continuous NE infusion	60 mcg/mL; clinician-titrated

A dose of 10 mcg per bolus at a concentration of 10 mcg/mL (equivalent to 1 mL per bolus) via peripheral intravenous access was consistent across all three included studies.

3.4 Efficacy Outcomes

3.4.1 Push-Dose NE versus Ephedrine (Hassani et al. 2018)

Hassani et al. (2018) randomised 56 hypertensive patients undergoing spinal surgery to either push-dose NE 10 mcg or ephedrine 5 mg bolus for IOH management under general anaesthesia. Push-dose NE demonstrated clear superiority across multiple efficacy parameters (Table 3).

Table 3. Efficacy Outcomes: Push-Dose NE vs. Ephedrine (Hassani et al. 2018, n=56)

Outcome Measure	NE Bolus (n=28)	Ephedrine (n=28)	p-value
MAP at end of anaesthesia, mmHg (mean ± SD)	98.85 ± 12.65	91.48 ± 10.93	<0.001
MAP at 5 min after first hypotension episode	Higher	Lower	<0.001
Number of hypotension episodes (mean)	Fewer	More	0.005
Number of vasopressor doses per episode	Fewer	More	<0.001
Total vasopressor dose during anaesthesia	Fewer	More	0.004
Heart rate at end of anaesthesia, bpm (mean ± SD)	70.5 ± 10.75	77.5 ± 9.22	0.012

Push-dose NE significantly elevated MAP compared to ephedrine at the end of anaesthesia (98.85 vs. 91.48 mmHg, $p<0.001$). The NE group experienced fewer hypotension episodes ($p=0.005$) and required fewer total vasopressor doses ($p=0.004$). Heart rate was significantly lower in the NE group (70.5 vs. 77.5 bpm, $p=0.012$), consistent with the pharmacological profile of norepinephrine, which lacks direct chronotropic activity.

3.4.2 Push-Dose NE versus Continuous NE Infusion: Low-to-Moderate Risk Patients (Thomsen et al. 2026 / INDUCT Trial)

The INDUCT trial randomised 261 low-to-moderate risk patients (ASA ≥II, age ≥45 years) to NE bolus or continuous NE infusion for 15 minutes following induction of anaesthesia. The primary outcome was the area under the MAP <65 mmHg curve (AUC-hypo) over 15 minutes,

measured by a blinded continuous non-invasive arterial pressure (CNAP) system.

Table 4. Efficacy Outcomes: Push-Dose NE vs. Continuous NE Infusion in Low-to-Moderate Risk Patients (Thomsen et al. 2026, n=261)

Outcome Measure	NE Bolus (n=131)	Continuous NE Infusion (n=130)	p-value
AUC MAP <65, mmHg·min (median, IQR)	5.5 (0.5–24.5)	3.6 (0.0–16.6)	0.070 (NS)
Duration MAP <65, min (median, IQR)	1.4 (0.2–3.2)	1.0 (0.0–2.5)	0.052 (NS)
Patients experiencing ≥1 MAP <65 episode (%)	Higher	Lower	0.022 (S)
Total NE dose, mcg/kg (median, IQR)	0.3 (0.2–0.4)	0.9 (0.6–1.1)	<0.001

In low-to-moderate risk patients, NE bolus showed **comparable hypotension burden** to continuous NE infusion, with no statistically significant difference in the primary outcome (AUC-hypo, p=0.070) or duration of MAP <65 mmHg (p=0.052). However, the proportion of patients experiencing at least one MAP <65 episode was higher in the bolus group (p=0.022). Notably, total NE consumption was significantly lower with the bolus approach (0.3 vs. 0.9 mcg/kg, p<0.001), suggesting that the bolus strategy achieves comparable haemodynamic control with substantially less drug.

3.4.3 Push-Dose NE versus Continuous NE Infusion: High-Risk Patients (Vokuhl et al. 2025)

Vokuhl et al. (2025) randomised 71 high-risk patients (ASA II–IV, age ≥45 years) undergoing elective major noncardiac surgery to NE bolus or continuous NE infusion, with continuous invasive arterial monitoring (radial catheter). The primary outcome was average real variability of MAP (ARV-MAP), a measure of blood pressure variability reflecting haemodynamic stability.

Table 5. Efficacy Outcomes: Push-Dose NE vs. Continuous NE Infusion in High-Risk Patients (Vokuhl et al. 2025, n=71)

Outcome Measure	NE Bolus (n=35)	Continuous NE Infusion (n=36)	p-value	Effect Size
ARV-MAP, mmHg/min (mean ± SD) [Primary]	25 ± 7	19 ± 6	<0.001	Cohen's d=1.00
AUC MAP <65, mmHg·min (median, IQR)	9 (1–16)	3 (0–18)	0.178 (NS)	Cliff's d=0.19
Duration MAP <65, min (median, IQR)	1.8 (0.5–2.8)	0.8 (0.0–2.4)	0.027 (S)	Cliff's d=0.30
Total NE dose, mcg/kg (median, IQR)	0.5 (0.4–0.7)	1.3 (1.0–1.7)	<0.001	Cliff's d=–0.87

In high-risk patients with continuous arterial monitoring, **continuous NE infusion provided significantly superior haemodynamic stability** compared to NE bolus, as evidenced by a

significantly lower ARV-MAP (19±6 vs. 25±7 mmHg/min; $p<0.001$; Cohen's $d=1.00$, indicating a large effect). The duration of MAP <65 was also shorter with continuous infusion (0.8 vs. 1.8 min; $p=0.027$). However, the area under MAP <65 (the measure used in the INDUCT trial) did not differ significantly ($p=0.178$). Total NE dose remained considerably lower with the bolus regimen (0.5 vs. 1.3 mcg/kg; $p<0.001$).

3.5 Safety Outcomes

3.5.1 Overshoot Hypertension

Table 6. Overshoot Hypertension: Push-Dose NE vs. Comparator

Study	Definition	NE Bolus	Comparator	p-value
Hassani (2018)	Not reported	Not reported	Not reported	N/A
Thomsen (2026)	AUC above MAP >100 mmHg·min	42.0 (18.2–86.9) mmHg·min	47.4 (8.7–131.1) mmHg·min (continuous)	0.390 (NS)
Vokuhl (2025)	AUC above MAP >100 mmHg·min	16 (6–53) mmHg·min	20 (2–72) mmHg·min (continuous)	0.986 (NS)

No statistically significant differences in overshoot hypertension were observed between NE bolus and continuous NE infusion in either the INDUCT trial ($p=0.390$) or the Vokuhl study ($p=0.986$).

3.5.2 Arrhythmia

No study in this systematic review specifically included arrhythmia as a structured, pre-specified outcome measure. Hassani et al. (2018) excluded patients with known arrhythmia history from enrolment. The higher heart rate observed in the ephedrine group (77.5 vs. 70.5 bpm, $p=0.012$) in that study suggests a sympathomimetic tachycardia effect of ephedrine, but formal arrhythmia events were not quantified. Neither Thomsen et al. (2026) nor Vokuhl et al. (2025) reported any arrhythmia adverse events.

3.5.3 Peripheral Intravenous Access Complications

No vascular access complications related to peripheral IV administration of push-dose NE were reported in any of the three included studies.

3.6 Heterogeneity Assessment

Marked clinical and methodological heterogeneity was identified across the three included studies (Table 7). The primary outcomes differed substantially: Hassani et al. used absolute MAP values; Thomsen et al. used AUC-hypo (MAP <65 mmHg); and Vokuhl et al. used ARV-MAP. Threshold definitions for hypotension also differed (MAP <60 mmHg in ART-001 vs. MAP <65 mmHg in ART-002 and ART-005). The comparator groups were fundamentally different (ephedrine bolus in ART-001; continuous NE infusion in ART-002 and ART-005). Patient risk profiles ranged from ASA I–II hypertensive patients (ART-001) to high-risk ASA II–IV patients (ART-005). These factors precluded pooled meta-analytic synthesis.

Table 7. Sources of Clinical and Methodological Heterogeneity

Dimension	ART-001 (Hassani 2018)	ART-002 (Thomsen 2026)	ART-005 (Vokuhl 2025)
Hypotension definition	MAP <60 mmHg	MAP <65 mmHg	MAP <65 mmHg
Comparator	Ephedrine bolus	Continuous NE infusion	Continuous NE infusion
BP monitoring modality	Invasive (arterial)	Intermittent oscillometric + blinded CNAP	Invasive (arterial)
Patient risk profile	Hypertensive, ASA I–II	Low-moderate risk, ASA ≥II	High-risk, ASA II–IV
Primary outcome	MAP at end of anaesthesia; hypotension episodes	AUC MAP <65 mmHg	ARV-MAP
Observation window	Entire anaesthesia	15 min post-induction	15 min post-induction
Country	Iran	Germany	Germany

3.7 Risk of Bias Assessment

Risk of bias was assessed using the Cochrane RoB 2 tool across five domains (Table 8). All three studies received an overall judgement of "Some concerns," with no study rated as "High risk."

Table 8. Risk of Bias Summary (Cochrane RoB 2)

Study	D1: Randomization	D2: Deviation from Intervention	D3: Missing Data	D4: Outcome Measurement	D5: Selective Reporting	Overall
Hassani et al. (2018)	Some concerns ¹	Low	Low	Low	Low	Some concerns
Thomsen et al. (2026)	Low	Some concerns ²	Low	Low	Low	Some concerns
Vokuhl et al. (2025)	Low	Some concerns ²	Low	Low	Low	Some concerns

¹ Allocation concealment not explicitly described; table-of-random-numbers method used but potential predictability cannot be excluded.

² Anaesthesiologists administering the intervention could not be blinded to bolus vs. infusion mode due to the nature of the intervention; however, primary outcomes were measured objectively (CNAP, arterial catheter data) or algorithmically (ARV-MAP).

Domain D3 (missing outcome data): Hassani—100% data completeness; Thomsen—94.5% (261/276); Vokuhl—98.6% (71/72). Domain D5 (selective reporting): all studies had pre-registered protocols (IRCT, NCT05940649, NCT05997303).

The "some concerns" ratings in D2 for Thomsen and Vokuhl represent an inherent and unavoidable limitation of all studies comparing mode of administration (bolus vs. infusion), where provider-level blinding is not feasible. The impact on outcome measurement was mitigated by objective, algorithmically-derived outcome measures.

3.8 Certainty of Evidence (GRADE)

Table 9. GRADE Evidence Certainty Assessment

Outcome	Studies (n)	Study Design	Risk of Bias	Inconsistency	Indirectness	Imprecision	Evidence Certainty
Hypotension reversal (MAP target)	3 RCTs	RCT	Downgrade 1: Some concerns in all studies	Downgrade 1: Different outcomes, mixed direction	Not downgraded	Downgrade 1: Small total n (388)	⊕⊕■ Low
Haemodynamic stability (ARV-MAP)	1 RCT	RCT	Some concerns	N/A	Indirect (high-risk only)	Downgrade 1: Single study	⊕⊕■ Low
Overshoot hypertension	2 RCTs	RCT	Some concerns	Not downgraded	Not downgraded	Downgrade 1: Imprecision (wide IQR)	⊕⊕■ Low
Arrhythmia	0 studies	—	—	—	—	—	No evidence

The overall certainty of evidence was **Low** (⊕⊕■), primarily due to: (1) significant clinical heterogeneity across studies; (2) limited total sample size (n=388); and (3) the inherent risk of bias associated with unblinded vasopressor administration.

4. Discussion

4.1 Summary of Findings

This systematic review identified three RCTs evaluating push-dose norepinephrine for intraoperative hypotension in non-obstetric surgery. Despite the limited number of studies and the inability to perform meta-analysis, several important observations emerge:

1. Push-dose NE 10 mcg at a concentration of 10 mcg/mL via peripheral IV is the **consistently used dose and preparation** across all studies.
2. **Compared to ephedrine**, push-dose NE is superior in maintaining MAP and reducing the total vasopressor burden in hypertensive patients undergoing spinal surgery under general anaesthesia.
3. **Compared to continuous NE infusion in low-to-moderate risk patients**, push-dose NE provides comparable haemodynamic control (measured by AUC-hypo) with significantly less total drug consumption.
4. **In high-risk patients** with continuous arterial monitoring, continuous NE infusion provides superior blood pressure stability (lower ARV-MAP), suggesting that for this patient subgroup, continuous infusion may be the preferred strategy.
5. The **safety profile** of push-dose NE via peripheral IV is favourable: no significant overshoot hypertension compared to comparators, no arrhythmia events documented, and no vascular complications reported.

4.2 Push-Dose NE versus Ephedrine

The superiority of push-dose NE over ephedrine observed by Hassani et al. (2018) is consistent with pharmacological principles. Ephedrine acts primarily through indirect sympathomimetic mechanisms, displacing norepinephrine from presynaptic terminals; this indirect mechanism results in slower onset, greater heart rate elevation, and greater variability in response.²¹ Norepinephrine, acting directly on alpha-1 adrenergic receptors, produces a more predictable and potent vasopressor response with minimal chronotropic effect.

The lower heart rate in the NE group (70.5 vs. 77.5 bpm, $p=0.012$) is clinically significant, as tachycardia increases myocardial oxygen demand—a particular concern in the surgical patient with potential coronary artery disease. These findings align with the evolving body of evidence in obstetric anaesthesia that has demonstrated the superiority of phenylephrine and norepinephrine over ephedrine for spinal hypotension, primarily due to the avoidance of beta-adrenergic-mediated adverse effects.^{22,23}

4.3 Push-Dose NE versus Continuous NE Infusion: Risk Stratification Matters

The differing results between the INDUCT trial (Thomsen et al. 2026) and Vokuhl et al. (2025) highlight the critical importance of **patient risk stratification and blood pressure monitoring modality** in determining the optimal vasopressor strategy.

In the INDUCT trial (low-to-moderate risk patients, intermittent BP monitoring), bolus NE was non-inferior to continuous infusion for the primary outcome of AUC-hypo. This suggests that for the broad population of patients undergoing elective non-cardiac surgery with standard monitoring, a reactive push-dose strategy is adequate.

In contrast, among high-risk patients monitored with invasive continuous arterial pressure (Vokuhl 2025), continuous NE infusion resulted in significantly lower blood pressure variability (ARV-MAP), a metric associated with improved outcomes in critically ill patients.^{24,25} The large effect size (Cohen's $d=1.00$) is clinically meaningful and suggests that the bolus strategy—inherently reactive rather than proactive—generates greater beat-to-beat MAP fluctuation when applied in patients with more labile haemodynamics.

This divergence underscores the concept that haemodynamic management should be **individualised** based on patient risk, surgical complexity, and available monitoring. For high-risk patients, or those in whom blood pressure variability carries heightened clinical risk (e.g., patients with severe coronary artery disease, carotid stenosis, or recent stroke), a proactive continuous infusion approach may be preferable.

4.4 Safety Considerations

The use of norepinephrine via peripheral intravenous access has historically raised concerns about the risk of tissue necrosis from extravasation.²⁶ However, this risk appears to be dose- and concentration-dependent. At the low concentrations and doses used in push-dose NE protocols (10 mcg/mL, 10 mcg per bolus), this risk is substantially mitigated. None of the three included studies reported vascular or tissue complications related to peripheral administration, and a growing body of non-systematic evidence supports the safety of peripheral administration of low-concentration NE for short-term use in the perioperative setting.^{27,28}

The absence of overshoot hypertension difference compared to comparators ($p=0.390$ and $p=0.986$) is reassuring, though the lack of structured outcome measurement for hypertensive events in Hassani et al. (2018) limits conclusions across all comparators.

The absence of arrhythmia data across all three studies is a notable limitation. While the low heart rate in the NE group in Hassani et al. (2018) indirectly suggests absence of alpha-mediated reflex bradycardia at these doses, formal cardiac monitoring data were not systematically collected in any study. Future studies should include structured arrhythmia outcome measurement.

4.5 Clinical Implications

Based on the available evidence, the following clinical implications can be drawn:

1. **Push-dose NE 10 mcg** (10 mcg/mL, 1 mL bolus via peripheral IV) represents an effective and safe first-line vasopressor for acute IOH in non-obstetric surgery in low-to-moderate risk patients.
2. The bolus can be repeated every 2–5 minutes based on MAP response. The consistent use of a standardised concentration (10 mcg/mL) across studies supports this as a practical preparation for operating room use.
3. In **high-risk patients** (ASA $\geq$ III, major noncardiac surgery, significant cardiovascular comorbidities) with continuous arterial monitoring, a **continuous NE infusion** may be preferred for superior haemodynamic stability.
4. Push-dose NE is preferable over ephedrine when avoidance of tachycardia and chronotropic effects is clinically important, such as in patients with coronary artery disease or severe hypertension.
5. Peripheral IV administration at this concentration and dose appears safe in the short-term intraoperative setting, based on the current evidence base.

4.6 Limitations of this Review

1. **Limited number of included studies:** Only three RCTs met inclusion criteria, limiting generalisability.
2. **Clinical heterogeneity:** Differences in hypotension definitions (MAP <60 vs. <65 mmHg), comparators, primary outcome measures, patient populations, and monitoring modalities precluded pooled analysis.
3. **No meta-analysis:** The qualitative nature of the synthesis limits the strength of comparative conclusions.
4. **Absence of arrhythmia data:** None of the studies systematically measured arrhythmia as an outcome, leaving a critical safety gap.
5. **Limited population diversity:** ART-001 was conducted exclusively in hypertensive patients undergoing spinal surgery in Iran, limiting extrapolation. ART-002 and ART-005 focused on the first 15 minutes post-induction, not the full intraoperative period.
6. **Potential for publication bias:** With only three studies, assessment of publication bias using funnel plots was not feasible.

7. **Two potentially relevant reviews were excluded:** Berkenbush et al. (2024) and Duarte-Medrano et al. (2026) were excluded as they were review articles; primary studies cited within them may include additional relevant data not captured by our search.

4.7 Future Research Directions

1. **Multicentre RCTs** with standardised hypotension definitions (MAP <65 mmHg sustained for >1 minute, per POQI-3 consensus) ■ comparing push-dose NE vs. other vasopressors across diverse surgical populations and anaesthetic techniques.
 2. Studies specifically powered to detect differences in **arrhythmia, myocardial injury, and acute kidney injury** as primary safety outcomes.
 3. **Pharmacokinetic/pharmacodynamic studies** to characterise the dose-response relationship of push-dose NE and optimise dosing regimens for different patient subgroups.
 4. Research in **resource-limited settings** where push-dose NE via peripheral IV offers a practical, low-cost vasopressor option without requiring advanced monitoring infrastructure.
 5. Studies exploring **combination strategies** (e.g., pre-emptive push-dose NE vs. reactive dosing) and their impact on long-term perioperative outcomes.
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5. Conclusions

Push-dose norepinephrine 10 mcg per bolus at a concentration of 10 mcg/mL, administered via peripheral intravenous access, is an effective and safe vasopressor for the management of intraoperative hypotension in adult patients undergoing non-obstetric surgery. It demonstrates superiority over ephedrine in MAP maintenance and reduction of hypotension episodes, with a more favourable heart rate profile. In low-to-moderate risk patients, push-dose NE is comparable to continuous norepinephrine infusion in terms of hypotension burden, while requiring significantly less total drug. In high-risk patients with continuous arterial monitoring, continuous infusion provides superior haemodynamic stability and should be considered preferentially. The safety profile of peripheral push-dose NE at these doses is favourable based on current evidence, with no significant overshoot hypertension or adverse vascular events documented.

The overall certainty of evidence is Low (GRADE ⊕⊕■■■), reflecting the limited number of studies, clinical heterogeneity, and small total sample size. High-quality, multicentre RCTs with standardised outcome definitions and structured safety assessments are required to strengthen the evidence base and guide clinical practice.

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